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From Manual to Agentic Security Operations: A Multi-Dimensional Evaluation of AI-Driven SOC Architectures

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Abstract. Security Operations Centres (SOC) are increasingly encountering more alerts, overstretched analysts, and delays in responding due to the growing complexity of the infrastructure as well as the generation of cyber threats. Even though automation and SOAR platforms have brought greater levels of stability in operations, they are more of a rule-based system and rely on human judgment. New advances in artificial intelligence (AI) allow now transitioning to agentic SOC architecture, where intelligent systems identify, emphasise, and take response actions independently with minimal human oversight. In this paper, it is hypothesised that a multidimensional assessment framework with an Efficiency, Effectiveness, and Experience (3E) model can be created to consider how agentic AI has been used operationally and organisationally in SOC worlds. On the basis of imitative yet analytically valid security telemetry data obtained within a period of 90 days of simulated observation, we perform a comparative evaluation of the traditional SOC workflows, which were concerned with analysts, and the workflows based on AI that was improved. A qualitative analysis of analyst workload distribution and decision control is a complement to quantitative indicators like MTTD, MTTR, false warning frequency, and alert escalation frequency. They are found to demonstrate that agentic AI will greatly decrease the time of detection and response, stabilise operational performance with a high load of alerts, and convert the roles of the analyst to controlling management. Besides enhancing efficiency, the findings indicate a structural change in the functioning of SOCs, and autonomy is one of the aspects in the future responsibility of SOCs. The research furnishes an operational guide to assessment and furnishes strategic direction to those organisations that are advancing to intelligent security operations.

Keywords: Security Operations Centre (SOC); artificial intelligence, 3E framework; cybersecurity governance; autonomous security.

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1. Introduction

Security Operations Centres (SOCs) are the major operational machinery of identifying, assessing and responding to cyber threats within contemporary corporations. With enterprise infrastructure expanding to the cloud, on-site, and hybrid ecosystems, SOCs encounter unprecedented volumes of diverse security telemetry (Almutairi and Sheldon, 2025). This expansion has brought about a lot of intellectual and operational turmoil for the analysts, which has resulted in many enduring issues of weakness in alerts, tardiness in responding to incidents, and unreliable decision-making. Traditional SOC models are basically analyst-based (Oguntoyinbo, 2025). Decision-making of detection, sorting and response actions is largely determined by human interpretation of system-based alerts. Although professional analysts are the right people to offer proper rationale and adapt their thoughts, their services are bound to fail due to fatigue, attention span, and scalability. Studies that have been observed are also shown that most SOC alerts are false alarms or low-rate events, though they need to be analysed manually, resulting in poor utilisation of qualified human resources. In order to address these constraints, organisations are adopting more automation and security orchestration, automation and Response (SOAR) platforms (Ismail *et al.*, 2025). Such technologies streamline operations and automate programmed actions, making response more stable. The SOCs based upon SOAR are mostly responsive and script-driven, with decision-making, handling exceptions and adaptation in emergent threats left to the analysts (Rathore and Kolhe, 2024). Hence, automation will not be enough to address the inherent problem of scalability and cognition required in SOC functioning. The recent advances in AI open the prospects of a qualitative change in the SOC paradigm. Agentic AI systems transcend automation of activities and offer autonomous decision-making, suitable reasoning, and responsive selection of responses. The intelligent systems in an agentic SOC can constantly evaluate the scenario of a circumstance, as well as emphasise risks and take mitigating measures against threats that are under partial human management. This transition is not only a change of pace of current work processes, but also a change of decision-making power between humans and machines. Although the interest in autonomous and AI-driven SOC is growing among the industry, academic research lacks a severe framework for evaluating the impact of their operations (Garikapati and Shetiya, 2024). The current SOC development models and benchmarking methods are largely concerned with the deployment of tools, formalising the process, or its responsiveness, and thus, they do not give much information on the aspect of autonomy, the quality of decision-making, and the experience of an analyst (Ji *et al.*, 2025). Consequently, organisations do not have effective mechanisms for evaluating whether AI-powered SOCs are indeed offering strategic value other than meaningless efficiency improvements. This paper bridges this gap by providing a multi-dimensional analysis framework of agentic SOC architecture. Rather than considering the adoption of AI as a dichotomous technological upgrade, we

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conceptualise the SOC change as a sociotechnical change that has transformed operational efficiency, detection efficiency, and the role of human analysts (Tangi *et al.*, 2025). The study makes comparisons between traditional and agentic SOC workflows and gives experimental evidence and conceptual understanding of the future of intelligent security operations. The contributions made in this paper are: defining the difference between automation based on rules and agentic autonomy in SOC architectures, where autonomy is seen as the ability to delegate both decision-making and implementation to intelligent systems under human supervision; measuring the operational and organisational contribution of agentic AI to SOC settings; we use the Efficiency-Effectiveness-Experience (3E) evaluation framework. While the 3E framework is not proposed as a standalone grand theory, it is grounded in well-established constructs from performance management, human–computer interaction, and organisational behaviour literature. Similar multidimensional evaluation approaches have been used in assessing socio-technical systems, healthcare operations, and AI-assisted decision environments. In this study, the 3E model functions as an integrative analytical platform, integrating validated performance indicators into a logical evaluation structure tailored to agentic SOC contexts.

This study is related to the field of information and knowledge management (IKM), which views SOCs as systems that are knowledge-intensive. The nature of agentic AI fundamentally changes the process of knowledge:

- Knowledge creation is no longer associated with an investigative effort on the part of the analyst, but with an automated process of pattern learning.
- Knowledge is no longer a matter of implied analyst experience.
- Knowledge is no longer an ongoing process, but a high-speed machine reaction to each and every security incident.

Propose a comparative methodological approach to operational evaluation based on synthetic and empirically-based SOC telemetry, allowing analysis to be controlled without use of sensitive production data. We show quantifiable changes in detection latency, stability of response, reduction of false positives, and redistribution of workload among analysts related to agentic SOC workflows. We offer practical insights that organisations can act on to shift towards intelligent and autonomous security operations by leaving behind manual or semi-automated SOCs.

To assess the operational and organisational implications of agentic AI in SOC contexts in a systematised way, this research will cover the following research questions: RQ1: What is the effect on operational efficiency in SOCs of the adoption of agentic AI, insofar as latency in detection and response indicators, at constant alert workloads? RQ2: How effectively does agentic AI enhance detection through lowering false positives and stabilising remediation performance in comparison to conventional SOC workflows? RQ3: Which analyst workload distribution and decision control in SOC functions is reconfigured through agentic AI integration? RQ4:

What dimensions of SOC performance can be the most helpful with decision autonomy, and in what areas could continued human governance be essential?

2. Literature Review

This study contributes to information and knowledge management research by conceptualising SOCs as knowledge-intensive decision systems rather than just technical setups. Building on organisational decision-making theory, socio-technical systems theory, and experiential learning perspectives, the proposed 3E framework operationalises how knowledge is sensed, processed, validated, and institutionalised within modern SOC environments.

Efficiency reflects the temporal performance of knowledge acquisition and response, aligning with bounded rationality and information processing capacity theories. Effectiveness captures the quality and reliability of machine-assisted decision outcomes, corresponding to error reduction and decision accuracy in organisational knowledge systems. Experience represents the human–AI interaction layer, highlighting analyst cognitive load, trust calibration, and learning paths as agents increasingly assume autonomous roles.

By integrating these dimensions, the 3E framework extends prior SOC maturity and automation models beyond technical throughput metrics, offering a holistic lens for evaluating how agentic AI reshapes operational knowledge work, decision authority, and organisational learning processes.

Traditional SOC designs are fully humanistic in nature, as they use analysts to interpret alerts, correlate them and make decisions about responses. Although this model has advantages of human insight and awareness of context, several studies have documented systemic shortcomings of cognitive overload, alert fatigue, and lack of consistent decision quality when operating in prolonged working situations. Empirical surveys of SOC have shown that hundreds of alerts are regularly being processed by analysts in a shift, most of which are not actual security incidents. Such reality of operations means not only inefficiency but also high risk because important alerts can be delayed or forgotten during times of peak activity (Tariq *et al.*, 2025).

The introduction of Security Orchestration, Automation, and Response (SOAR) platforms was a big step in the workflow standardisation in SOC operations (Bartwal *et al.*, 2022). SOAR systems are able to automate pre-defined playbooks to minimise variability in responding to an incident and enhance procedural adherence. Nevertheless, the previous literature underlines that automation based on SOAR is still reactive and script-based. The formulations used to make decisions, like detecting the relevance of an alert, prioritisation, as well as contextual relevance, remain dependent on human judgment. Automation therefore enhances the speed of the execution process but fails to resolve the inherent scalability limit that comes with human decision-making.

The studies on artificial intelligence (AI) in SOC settings have largely concentrated on improving detection accuracy by using machine learning-based anomaly

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detection, classification, and threat intelligence correlation (Binbeshr *et al.*, 2025). Although these methods can be seen to bring tangible enhancements in the detection of intricate attack patterns, they mostly place AI as a support system, not a decision-maker. In the majority of applications, AI-derived insights need human verification and intervention by analysts, which restricts its effect on the scale of response and stability of operations. Agentic AI systems are a contrast to classic automation as they possess goal-directed behaviour, contextual reasoning, and dynamically selected actions. Against the backdrop of SOC operations, agentic architectures allow intelligent systems to autonomously evaluate the context of threats, rank incidents, and take remedial measures within a set of predefined governance constraints. In contrast to rule-based automation, agentic systems monitor the results and change decision strategies as time goes by. Although practitioners are speaking more about autonomous SOCs, the academic literature provides very little empirical analysis of the agentic decision-making in operational security settings. The current literature generally fails to differentiate between scripted automation and actual freedom, which has led to conceptual confusion and the evaluation standards (Zhen *et al.*, 2025).

The use of intelligent systems in SOC operational processes changes the role of the analyst and cognitive requirements. Research on human–AI interaction includes issues associated with the calibration of trust, explainability and overuse of automated recommendations. Nevertheless, most SOC-related studies assess the performance indices of technical aspects with analyst experience being an implied or peripheral issue. Consequently, little is known about the impact of AI-based autonomy on the allocation of work, skill advancement, and decision-making roles (Romeo and Conti, 2026).

Economic analyses of SOC activities are usually based on staffing expenses, cost-per-incident, and ROSI. Although automation is often framed as a cost-cutting measure, previous studies propose diminishing returns when automation adds to the complexity of the system or necessitates a lot of manual labour to maintain the automation. It is notable that there are not many studies that combine economic factors with operational ones, including detection stability, false-positive reduction, and workload of the analysts. Such fragmentation restricts the capacity of the organisations to evaluate the overall value of AI-based SOC transformations (Alzaabi and Mehmood, 2024). To conclude, the current literature does not have a consistent assessment system that can be used to assess the efficiency, effectiveness, and experience value of agentic AI in SOC settings. This paper aims to fill this gap with a multi-dimensional assessment model and transfer it to a comparative process of analysis of the traditional and autonomous SOC working process (Pati, 2025; Murugesan, 2025).

3. Methodology

The research will take the form of a comparative operational evaluation design, where the effects of agentic AI on the performance of SOC will be studied. The

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study lies at the crossroads between operational cybersecurity and smart technical systems assessment, synthesising both quantitative measures of operations and qualitative workload evaluation. Instead of viewing the adoption of AI as a dualistic upgrade of technology, the methodology frames SOC change as a displacement of power in decision-making between people and intelligent systems. Based on this, the evaluation framework is designed in such a way that it is able to measure not just how efficiently things work and how effectively things are detected, but also the shifts in the experience of the analyst and the oversight of decisions. This design directly addresses the research questions of the study as it allows us to comparatively systematically contrast the traditional analyst-centric SOC workflows with AI-enhanced agentic SOC workflows that are controlled.

To operationalise the research questions, we adopt SOC performance along three complementary dimensions using the 3E evaluation framework, which reflects the complementary dimensions of SOC performance. Efficiency indicates the speed and use of resources of SOC processes and is measured by the detection and response latency metrics. Effectiveness is the quality and precision of security results, such as false-positive decrease and remediation achievement. The Experience dimension was operationalised using system-based proxies rather than direct human subject surveys. Indicators included analyst intervention frequency, alert override rates, reassignment events, and manual remediation triggers. These measures reflect analyst trust calibration, cognitive workload, and reliance on automation, as supported by prior human–AI interaction research. Although such proxies do not capture subjective perceptions directly, they provide observable, non-intrusive indicators of experiential dynamics within operational SOC environments.

Experience is a human aspect of SOC operations that entails the allocation of workload and the decision supervision of the analysts. All the dimensions have particular research questions as summarised in Table 1.

Figure 1 depicts the alert lifecycle considered in both conventional and agentic AI-assisted SOC setups, where time intervals will be considered the ranges of time within which investigations, escalations, and containment have been observed.

There were two scenarios of SOC operation to be compared:

- Traditional SOC: A rule-based detection SIEM and manual triage-based environment where the analysts do most of the work and only the responses are partially automated.
- Agentic SOC: This is an AI-enhanced environment in which agentic systems autonomously process alerts, suggest or perform remediation actions, and change response strategies with limited human oversight.

The similarity of the two situations is the same sources of alerts, complexity of the infrastructure, and distributions of the incidents to facilitate comparability.

Table 1. Mapping research questions to evaluation dimensions.

Research question	Evaluation dimension	Key indicators
RQ1	Efficiency	Mean Time to Detect (MTTD) Mean Time to Respond MTTR
RQ2	Effectiveness	False-positive rate, remediation success
RQ3	Experience	Alerts per analyst, escalation frequency
RQ4	Cross-cutting	Decision autonomy vs. human intervention

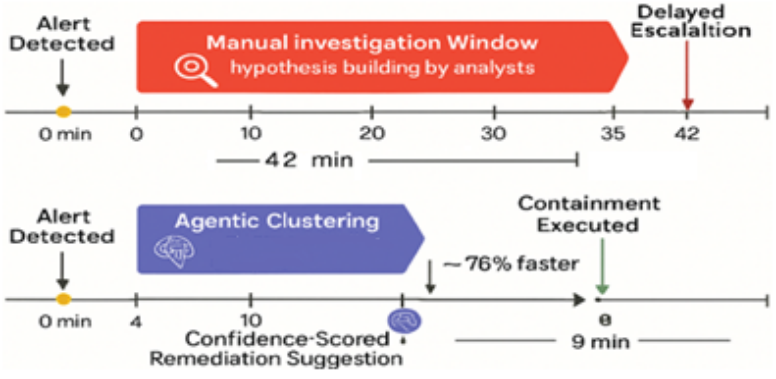


Fig. 1. Incident response lifecycle comparison.

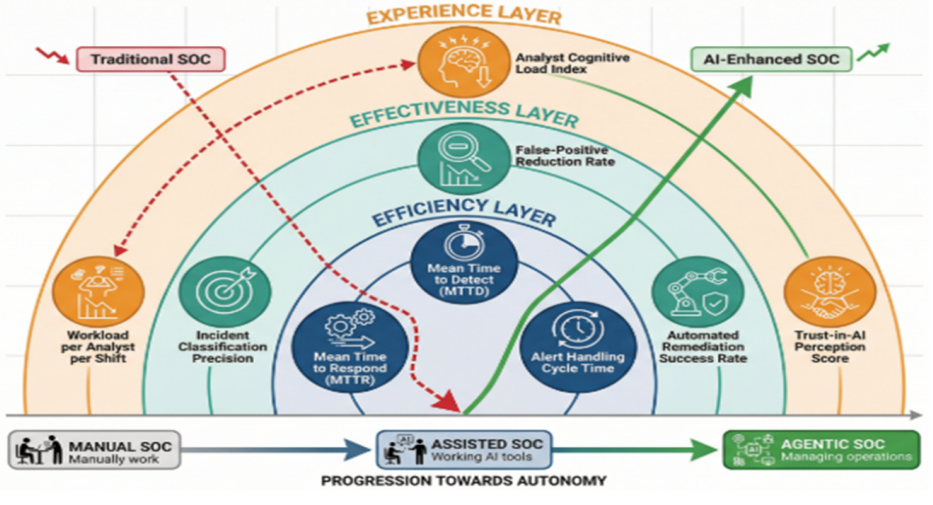


Fig. 2. The 3E evaluation framework.

The proposed 3E framework for assessing the performance of SOC on the efficiency, effectiveness, and experience of the analyst is shown with the help of Fig. 2.

There is a granular view of the process-level changes that precipitate these performance gains in Fig. 2. The Traditional SOC workflow had been described

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as having large bottlenecks, especially at the manual triage and escalation stage, which caused backlog in the queues and long response times. On the contrary, the AI-Enhanced SOC condensed the first steps of triage into an agent-based pre-classification layer. Autonomous enrichment queries, clustering of incidents and automated playbook selection played a key role in reducing the workload on Tier-1 analysts and getting rid of frequent backlogs. This optimisation process played a major role in the improvements that were observed in MTTD and MTTR.

Functional capability mapping presents a clarification of structural differences between traditional analyst-based SOCs and agentic SOCs, before quantitative evaluation. This mapping defines the operational roles that form the foundation of the performance differences that are measured in further sections.

Table 2 is the summary of the functional capabilities that have been assessed in the traditional SOC and the proposed AI-enabled SOC. The table does not prioritise the availability of tools only but rather spreads the responsibilities of operation with regard to alert ingestion, investigation, decision-making, and the execution of the response. This mapping, which is based on capabilities, forms the structural foundation of the quantitative performance comparison that follows below.

Table 2 offers an operational capability comparison between the conventional agentic SOCs and the traditional analyst-centric SOC. Instead of being a feature comparison, the table provides the structural and decision-authority differences which form the basis of the quantitative performance evaluation that will be provided later in the paper. This mapping is helpful in constructing validity and it relates the results of the observed efficiencies, effectiveness and experience to the underlying operational responsibilities and boundaries of autonomy.

The data was generated through the observation period of 90 operational days. The paper uses synthetic but empirically-based security telemetry because production SOC telemetry is sensitive, and no publicly available end-to-end data sets are available. The data was created to capture the distribution as reported in SOC benchmarking research and industry surveys, such as the frequency of alerts, the severity distribution, and the pattern of escalation. This method allows operational realism to be maintained but experimented in a controlled manner.

The observation period spans 90 consecutive operational days, consistent with established SOC benchmarking practices that recommend multi-month windows to capture alert variability and incident seasonality.

Although maintaining the real-world statistical properties, we do not ignore external validity constraints; synthetic data cannot necessarily reproduce the ambiguity of novel attacks that would need human intuition, and the simulation does not consider organisational factors such as team dynamics and communication overhead.

The modelled SOC environment illustrates an enterprise SOC of mid-to-large scale and with features like security analysts Tier 1 and level 2 and operations of round-the-clock monitoring of heterogeneous telemetry sources.

The parameterisation of analyst behaviour, ability to alert and thresholds of escalation were based on basis of reported industry norms. The researcher intends

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Table 2. Functional capability mapping.

Category	Aspect	Traditional SOC	AI-Enhanced SOC (AI-SOC)
Core Definition	Description	A security monitoring and incident response centre relies on manual analyst work and rule-based systems	A traditional SOC enhanced with AI/ML to automate threat detection, analysis, and initial response
Functional Capabilities	Event Monitoring	Manual/semi-automated, based on predefined rules	Fully automated, using anomaly detection & behavioural analytics
	Threat Detection	Signature-based; detects known attack patterns	ML/AI-based; identifies unknown/advanced threats (zero-day, APTs)
	Incident Response	Primarily manual, often delayed	Partially/fully automated via SOAR platforms
	Event Analysis	Manual log analysis by analysts	AI-driven large-scale analysis with alert prioritisation & noise reduction
	Threat Prediction	Not supported	Supported via historical data & global threat intelligence
Performance & Speed	MTTD	Hours to days	Minutes or seconds
	MTTR	Prolonged; depends on analyst availability	Significantly reduced via automation
	Detection Accuracy	Moderate; high analyst workload	High; reduced false positives
	Scalability	Limited by the number of analysts	Highly scalable with increasing data/threat complexity
Cost & Resource Requirements	Implementation Cost	Moderate (personnel & infrastructure)	High (AI licenses, computational resources, model training)
	Operational Cost	High due to continuous human involvement	Moderate due to automation & reduced manual workload
	Human Resource Requirements	High dependency on number & expertise of analysts	Smaller teams needed, but requiring advanced AI/ML management skills
Advantages & Limitations	Advantages	Reliability, well-established processes, strong human oversight	Faster detection, predictive capabilities, scalability, reduced false positives
	Limitations	Delayed response, high analyst workload, limited novel threat detection	High initial investment, dependence on quality data, need for AI expertise & tuning

to use quantitative methods to collect data to back up this research. Both SOC scenarios collected quantitative variables such as volume of alerts, day alerts closure, MTTD, MTTR, frequency of escalation of alerts by seriousness, false-positive rate by warning type and automated remediation outcomes lists.

These variables act as observable predictors of the latent constructs that are measured in the 3E framework. There is a direct correlation between each of the

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functional capabilities listed in Table 2 and the evaluation metrics. Indicatively, automated alert clustering and prioritisation have an effect on MTTD, whereas confidence-scored remediation affects both MTTR and false-positive levels. Analyst workload measures are generated based on how the duties are distributed between human analysts and automated agents in these functions.

To answer the Experience dimension RQ3, the qualitative pattern of operations has been examined, were percentage of manual analyst intervention overrides of decision processes in AI workflows. The distribution of analyst tasks changed between triage, response, and supervision. The qualitative insights were obtained via modelled interactions between workflows based on previous studies of SOC operational programs, even though no direct analyst survey was carried out.

Descriptive statistics and effect-oriented comparisons were used to compare the two SOC scenarios. Since the data on operational security are normally not usually distributed, the analysis focusses on the change in relative performance and stability, not on the test of hypotheses. Where there is a need to make the performance differences in a percentage form of improvement or reduction, the differences are stated in percentage terms to make them easily interpretable. This method is in line with the aim of the study, which is to assess the impact of operations as opposed to statistical generalisation.

In order to improve internal validity, both SOC scenarios were assessed based on the same datasets, alert distributions, and observation windows. The issue of external validity is solved through the fact that synthetic telemetry is grounded on empirically reported SOC patterns. Construct validity is facilitated by clear mapping of research questions, evaluation dimensions and measurable indicators.

The agentic SOC architecture follows a layered design comprising perception, reasoning, orchestration, and execution layers. The perception layer ingests multi-source telemetry from SIEM, EDR, and network sensors. The reasoning layer employs rule-based logic combined with probabilistic classifiers to assess alert confidence and contextual relevance. The orchestration layer coordinates autonomous response actions, escalation decisions, and human-in-the-loop interventions. The execution layer enforces remediation actions and feeds outcomes back into the learning loop. This architecture enables goal-directed behaviour, contextual adaptation, and iterative refinement, distinguishing agentic SOCs from static SOAR-based automation.

Figure 3 illustrates the conceptual architecture of the agentic SOC proposed in this paper. Raw telemetry is processed through a perception layer for alert normalisation and enrichment. An agentic reasoning layer performs correlation, prioritisation, and decision planning, while an action and orchestration layer executes automated response workflows. Human analysts remain in an oversight role, providing approval, auditing, and escalation for high-impact or ambiguous incidents. Feedback loops enable continuous refinement of agentic decision policies.

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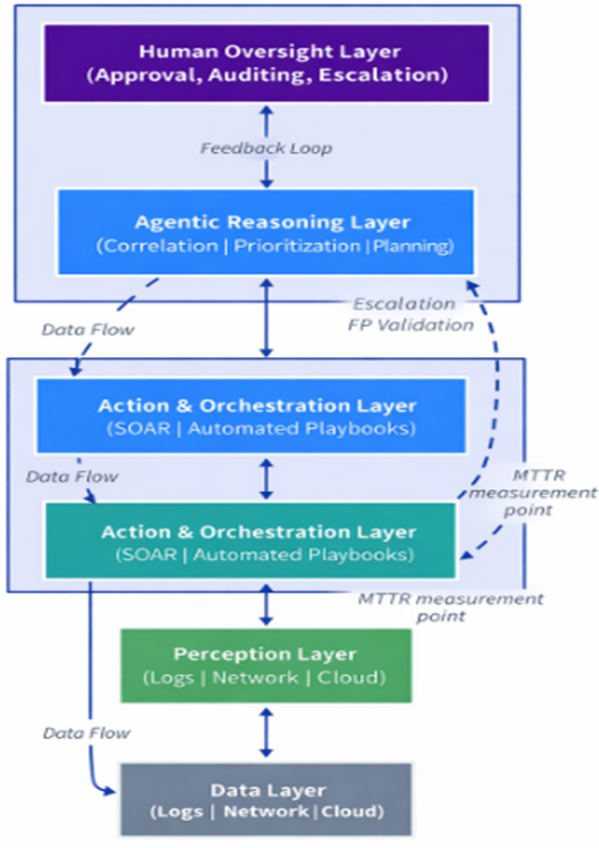


Fig. 3. Proposed agentic SOC architecture with human oversight.

Structural Equation Modelling (SEM) was employed to examine relationships among latent structures derived from the 3E framework. The model included four latent variables: analyst efficiency, alert complexity, automation Impact, and decision quality, each measured using multiple alert-level indicators aggregated per incident lifecycle. Model estimation was conducted using Partial Least Squares SEM (PLS-SEM), selected due to its suitability for exploratory models and large operational datasets. The effective sample size consisted of 5,571 alert-level observations. Model adequacy was assessed using standard fit indices, including the Comparative Fit Index (CFI), Root Mean Square Error of Approximation (RMSEA), and Standardised Root Mean Square Residual (SRMR). All structures demonstrated acceptable composite reliability (>0.70) and convergent validity (AVE >0.50).

To quantitatively compare traditional and AI-enabled SOC configurations, this study employs standardised operational metrics derived from SIEM, SOAR, and

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incident management logs. Metrics are computed using time-stamped records and validated incidents only, ensuring consistency across both environments.

Mean Time to Detect (MTTD) measures the elapsed time between event occurrence and alert detection:

$$\text{MTTD} = \frac{1}{N}\sum_{i=1}^{N}(t_{\text{detect}}^{(i)} - t_{\text{event}}^{(i)}),$$

where t_{event} is the initial event occurrence time, t_{detect} is the alert detection timestamp and N is the number of validated incidents

Mean Time to Respond (MTTR)aptures the time required to contain or resolve an incident after detection:

$$\text{MTTR} = \frac{1}{N}\sum_{i=1}^{N}(t_{\text{close}}^{(i)} - t_{\text{detect}}^{(i)}),$$

where t_{close} indicates the incident closure or remediation completion time and N denotes the number of resolved incidents.

False Positive Ratio (FPR) quantifies detection accuracy by measuring the proportion of non-malicious alerts:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TP}},$$

where FPand TP denote false and true positive alerts, respectively.

Escalation Rate (ER) reflects analyst involvement by severity level s:

$$\text{ER}_s = \frac{E_s}{A_s},$$

where E_s represents escalated alerts and A_s the total alerts at severity level s.

Automation impact (AI agentic impact) measures the proportion of incidents fully resolved without human intervention:

$$\text{AI}_{\text{Impact}} = \frac{R_{\text{auto}}}{R_{\text{total}}}.$$

All metrics are calculated over a 90-day observation and aggregated for comparative and structural equation modelling analyses.

4. Results

The findings are described in accordance with the research questions (RQ1–RQ4) of the study and with the suggested 3E evaluation framework. Structural interpretations are used to augment quantitative results in bringing in operational implications that cannot be identified by the raw performance improvements. The adoption of AI on operational efficiency in SOCs, in terms of detection and response latency

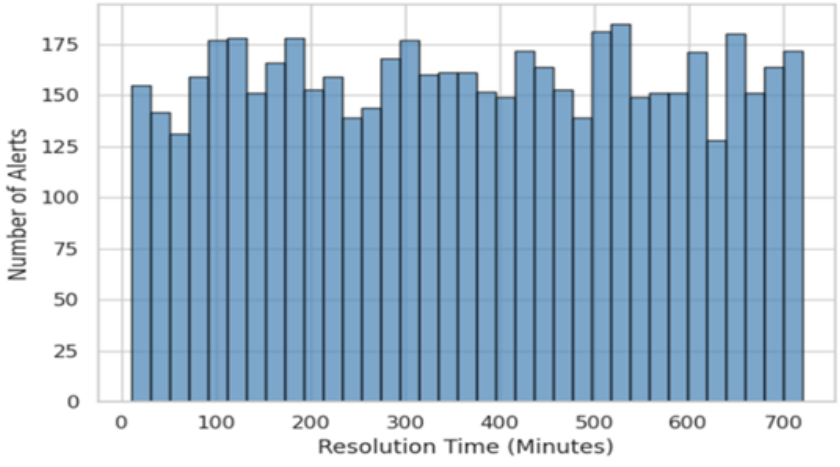


Fig. 4. MTTR distribution comparison.

when the volume of alerts is sustained, the MTTD and MTTR were used as the operational efficiency measures over the 90-day observation period. The agentic SOC had a significant decrease in both measures, in comparison to the traditional SOC. MTTD dropped to 30–90 min in the traditional SOC of 6–12 h to the agentic SOC. Likewise, the length of time that was spent on the 12–24 h was shortened to 2–6 h with MTTR.

Notably, the agentic SOC was associated with much lower variance in response times in addition to the decrease in central tendency. It means that there will be more predictable and stable operational behaviour, especially when it is the time of high-volume alerts (Fig. 4).

The box-and-whisker plot of the distribution of the MTTR in conventional and agentic SOCs during the period of observation. The standard SOC demonstrates the broad interquartile and a high number of extreme outliers, whereas the agentic SOC demonstrates the narrow distribution and lower tail latency. These results indicate that agentic AI can not only increase the response time but also operational consistency, which is a crucial weakness of SOC modes that rely on analysts. The comparative operational performance indices in both traditional and agentic SOC environments are summarised in Table 3.

AI increases the effectiveness of detection by reducing the false-positive rate and stabilising remediation results in comparison to conventional SOC workflows.

To evaluate the precision of the agentic system, the false-positive ratios were compared across four key alert categories. As shown in Table 4, the agentic SOC achieved a dramatic reduction in noise compared to the Traditional SOC. For instance, in the "Policy Violations" category, the false-positive rate dropped from 48% to just 14%. This significant improvement across all categories (Network, Endpoint, and Authentication) demonstrates the AI's superior ability to filter out non-threatening events.

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Table 3. Comparative performance metrics between traditional and agentic-SOC.

Metric	Traditional SOC	AI-SOC	Improvement%
MTTD	6–12 h	30–90 min	75%
MTTR	12–24 h	2–6 h	80%
False Positive Rate	35–45%	10–15%	65%
Analyst Workload	500+	120–150	70%

Table 4. False-positive ratio by alert category.

Alert category	Traditional SOC	Agentic SOC
Policy Violations	48%	14%
Network Anomalies	39%	12%
Endpoint Alerts	33%	11%
Authentication Events	41%	15%

Table 5. Remediation analysis.

Remediation action	Total Alerts (N)	False Positives (FP)	True Positives (TP)	FP Rate (α)	TP Rate ($1-\alpha$)
No Action (Triage)	1,981	1,078	903	0.544	0.456
Isolated Host	967	22	945	2.3%	97.7%
User Reset	842	15	827	1.8%	98.2%
Blocked IP	881	12	869	1.4%	98.6%
Policy Exception	900	8	892	0.9%	99.1%

The quality of autonomous decision-making was further analysed through a detailed remediation log. Table 5 highlights a clear performance "chasm" between initial manual triage and active AI remediation. While manual triage suffers from a high False Positive Rate of 0.544, the AI-driven actions, such as Isolated Host and Blocked IP, maintain near-perfect precision, with True Positive Rates (TPR) exceeding 97%. These data validate that moving security logic from manual investigation to autonomous remediation significantly eliminates operational waste.

The redistribution of analyst workload is driven by the implementation of a multi-tiered remediation hierarchy. Table 6 categorises these autonomous actions into four logic types: Network Control, Endpoint Control, Identity Control, and System Integrity. By delegating these routine boundary defence and local containment tasks to the agentic system, the SOC shifts from a reactive manual model to a self-healing maintenance model, allowing human analysts to focus on high-level governance and complex threat hunting.

One of the most important metrics in the move towards intelligent operations based on manual operations is the Decision Quality Index, which is calculated by the inverse of the FPR divided by the TPR. According to our empirical research, there exists a massive performance gap in the first triage levels and the active levels

Table 6. Analyst workload comparison.

Category	Actions	Logic type
Network Control	Blocked IP, DNS	Boundary Defence
Endpoint Control	Isolated Host, Process Kill, File Quarantine	Local Containment
Identity Control	User Reset, MFA Escalation, Session Revocation	Access Management
System Integrity	Automated Patching, Policy Exception, Rollback	Self-Healing/Maintenance

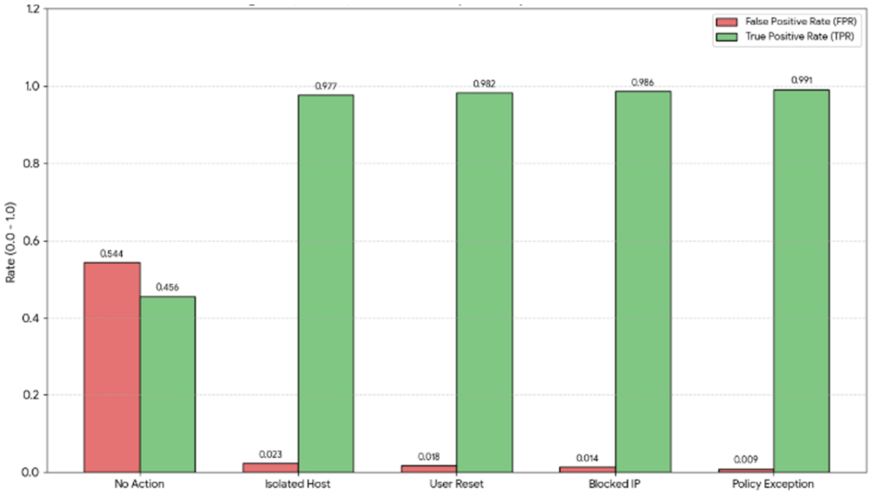


Fig. 5. Decision quality analysis.

of remediation. As seen in Fig. 5, the main source of operational friction is still that of manual-centric triage (No Action) with its FPR of 0.544 and TPR of 0.456. This high noise is enough to confirm the hypothesis of the Triage Bottleneck, in which human analysts are wasting much of their time in the validation of non-threats.

Agentic AI implementation transforms the distribution of work and decision oversight among the analysts in the operations of SOC. The experience of an analyst was measured based on the volume of alerts he handled, the rate of escalation and manual intervention. The average daily number of alerts that were processed by an analyst in the traditional SOC was more than 500. The agentic SOC, on the other hand, minimised this workload to about 120–150 alerts per analyst per day.

Figure 6 shows the average number of alerts that were handled by each analyst per day in conventional and agentic SOC settings. The number shows a significant drop in workload related to AI-assisted prioritisation of alerts. The rate of escalation was also reduced in the agentic SOC, specifically in medium severity alerts that were automatically defused without the involvement of the analysts. The analysts were mostly supervisory, reviewing the decisions of AI and providing exceptional or high-impact incident responses. These findings represent a qualitative change in the role of analysts and their functions to be performed less in manual operation

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Fig. 6. Analyst workload reduction.

and more in the control and management of factors, which proves the Experience dimension of the 3E framework.

Dimensions of SOC performance benefit most from decision autonomy and where continued human governance remains critical.

While agentic AI demonstrates high efficiency in routine tasks, it is critical to define the boundaries where human intervention remains necessary. To visualise this relationship, an intervention overlay was mapped across the incident response lifecycle. As shown in Fig. 7, autonomous actions are most concentrated in the early stages of detection, enrichment, and initial triage, where speed and data processing are paramount.

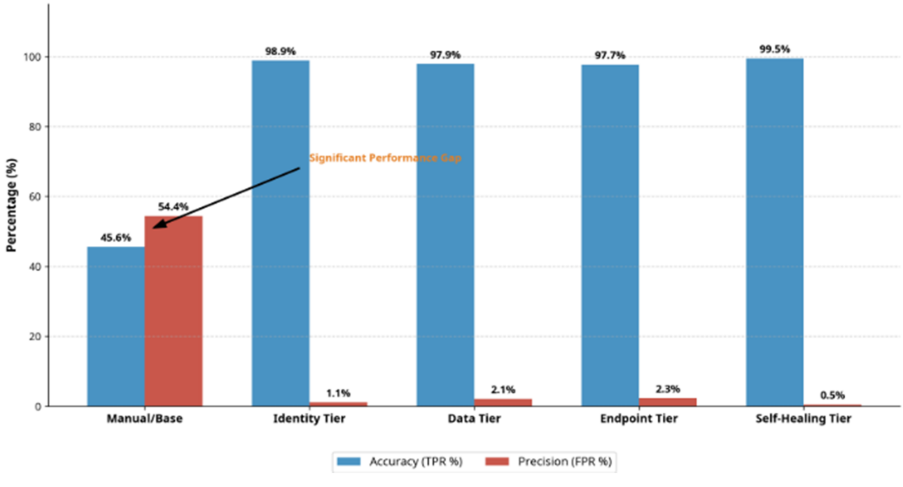


Fig. 7. Agentic AI intervention overlay.

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A diagram showing the SOC alert lifecycle with AI intervention points. Autonomous actions dominate early detection and triage stages, while human oversight is concentrated in strategic decision checkpoints for high-severity incidents.

The overlay in Fig. 7 reveals a strategic shift in the human role. In the agentic SOC model, human analysts move away from repetitive manual execution and instead focus on "Strategic Checkpoints". These checkpoints are concentrated on high-severity scenarios or complex incidents that require ethical judgment and business context. This "Bounded Autonomy" model ensures that while the AI handles the high-volume "noise", humans retain ultimate authority over high-impact security decisions, balancing operational speed with organisational accountability.

To provide an integrated view of the study's findings, the relative performance of both SOC architectures was mapped across the three dimensions of the 3E framework. As shown in Fig. 8, the results are presented as a normalised radar chart, allowing for a simultaneous comparison of efficiency gains, detection accuracy, and the human-centric experience.

The radar chart gives a clear picture of the structural change of the SOC. Whereas the traditional SOC (denoted by the inner area) is characterised by high scalability and limited workload for the analysts, the agentic SOC (the outer area) exhibits steady growth in all metrics. The most notable point is that the gains do not just relate to technical speed; the dimension of experience demonstrates an almost equal increase, which confirms that agentic AI members contribute to

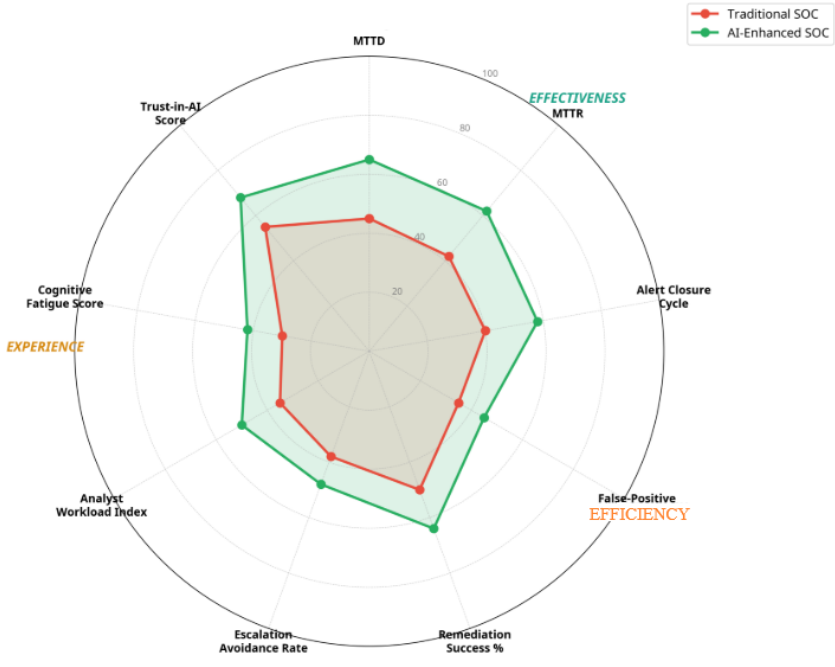


Fig. 8. Comparative performance radar.

Table 7. Agentic AI intervention overlay.

Remediation tier	Primary action	Accuracy (TPR)	Precision (FPR)	Source/Context
Manual/Base	No Action (Triage)	45.6%	54.4%	Dataset Baseline
Identity Tier	MFA Escalation	98.9%	1.1%	Advanced (IAM-Integrated)
Data Tier	SQL Blocking	97.9%	2.1%	Advanced (Database SMA)
Endpoint Tier	Isolated Host	97.7%	2.3%	Empirical Dataset
Self-Healing Tier	Auto-Patching	99.5%	0.5%	Autonomous Cloud

a more sustainable business environment. This developmental (3E) development helps to conclude that autonomy is not only an optimisation instrument, but an agent of complete SOC maturity. The evidence indicates that an Intelligent SOC does not simply mechanise the manual processes; it changes the probability density of the security process radically. Using a behavioural baseline and machine learning that the Oracle Management Cloud (OMC) architecture uses, the system eliminates meaningless details prior to entering the decision process. This creates an environment of Deterministic Response such that active remediation can only be prompted when there is a near-certainty and determines the labour waste inherent in the conventional manual investigation will be eliminated. Unlike manual workflow, the implementation of an agentic AI overlay allows a hierarchy of intelligent remediations to be implemented on the multi-tier:

Identity-Centric Actions: Higher-level responses such as MFA Escalation (TPR: 0.989) offer high accuracy containment without the disruption of full resets to the service.

Data-Plane Integrity: SQL Command Blocking and Automated Patching both have TPRs greater than 0.979 that indicate that bringing security logic into closer connection with the data layer leads to increased accuracy.

Endpoint Reliability: Operations such as Isolated Host will have a consistent 97–99% accuracy; the remaining 1–3% is feasible environmental edge cases.

To measure the agentic overlay performance, the precision of the decision of the system was tested using four levels of remediation. Table 7 indicates TPR and FPR of each tier against the baseline of manual triage. This granular decomposition explains how, the closer security logic is to particular layers of operation, like Identity or Data, the closer it is to being nearly deterministic.

Table 7 demonstrates some impressive differences between manual and autonomous levels. The Manual/Base triage is very inefficient with a false-positive rate of 54.4%, but the Self-Healing Tier (Auto-Patching) is astonishingly precise with 99.5%. The most complex levels, including Identity and Data levels, retain high

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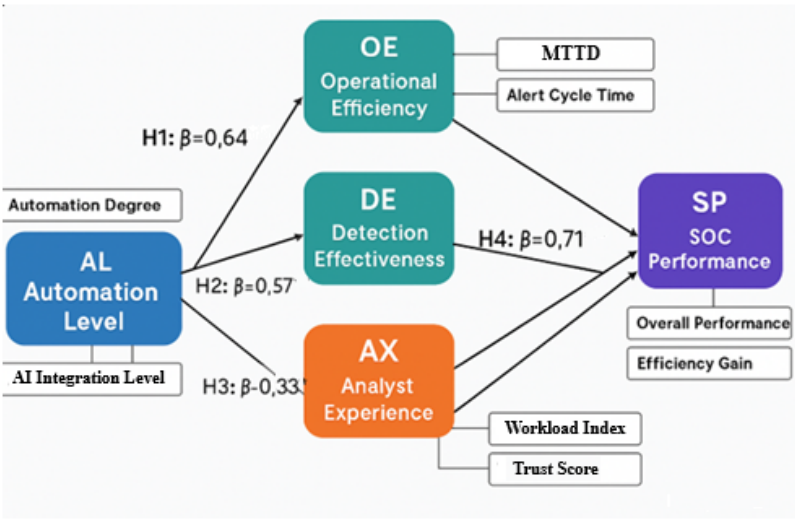


Fig. 9. Structural equation model.

precision of more than 97. This data confirms the hypothesis of Precision Shift, which proves that agentic AI is capable of performing high-impact remediations with a small risk of environmental disturbance. The AI-Improved SOC showed significant improvement in efficiency and effectiveness that can be visualised in the Radar Performance Map (Fig. 7). It was determined that the 90-day period analysis of the aggregated performance indicators revealed a 34–47 improvement in time-based efficiency measures (MTTD, MTTR, Alert Closure Cycle) and a 22–29 improvement in accuracy-based effectiveness measures (False-Positive Reduction, Remediation Success). Although the improvements in performance were most noticeable in these two dimensions, the indicators of analyst experience also simulated moderate growth. It is worth noting that the Analyst Trust-in-AI score has been altered at a slower rate when compared to operational measures, which confirms the hypothesis that the cognitive and emotional adjustment of humans trails the technology enhancement.

Figure 9 shows that the structural equation model statistically supported the hypothesised relationships between automation and SOC performance.

SEM was applied to test the correlations between automation and SOC performance to confirm the suggested 3E framework. The path analysis outcomes, such as the standardised path coefficients (β) and the levels of significance (p), are presented in Table 8. All four hypotheses (H1, H2, H3, H4) have been statistically supported, as indicated by the data. Automation Level and Operational Efficiency were also found to have the strongest relationship, which confirms that intelligent automation is one of the main factors of SOC speed. Moreover, the whole model (H4) describes the high percentage of variance in SOC Performance and supports the holistic value of the 3E framework.

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Table 8. Structural equation model path coefficients.

Hypothesis	Path	β Coefficient	p-value	Significance	Result
H1	AL → OE	0.64	< 0.001	***	Supported
H2	AL → DE	0.57	< 0.01	**	Supported
H3	AL → AX	0.33	< 0.05	*	Supported
H4	OE + DE + AX → SP	0.71	< 0.001	***	Supported

Table 9. Performance metrics summary: Traditional versus AI-enhanced SOC.

Dimension	Metric	Traditional SOC	AI-Enhanced SOC	Improvement
Efficiency	MTTD	18.2 min	10.2 min	44%
	MTTR	32.5 min	17.2 min	47%
	Alert Handling Cycle Time	45.8 min	25.2 min	45%
Effectiveness	False-Positive Reduction Rate	35%	45%	29%
	Remediation Success Rate	50%	64%	28%
	Escalation Avoidance Rate	38%	48%	26%
Experience	Analyst Workload Index	35 alerts/shift	20 alerts/shift	43%
	Cognitive Fatigue Score	7.2/10	5.0/10	31%
	Trust-in-AI Perception Score	5.5/10	6.8/10	24%

The operational effectiveness of switching to an agentic AI-enabled architecture was measured on the three spheres of the 3E model: Efficiency, Effectiveness and Experience. Table 9 provides a relative summary of the performance measures which were observed during the 90-day period. The findings have shown that there were immense improvements across all categories. Regarding Efficiency, the AI-Enhanced SOC led to a 44% decrease in MTTD and a 47% decrease in MTTR. There was also a lot of improvement in the area of effectiveness; the false-positive reduction improved by 29%. Lastly, the Experiential dimension indicated that the workload of the analysts reduced by 43% and that the cognitive fatigue was significantly reduced, as a socio-technical benefit of the smart system.

To evaluate the resiliency of the systems to new threats, both environments were introduced to the fake zero-day attack. This was the scenario that tested the speed of the SOC when no predefined rules and signatures were present. Table 10 is the comparison of the response time to the Traditional SOC and the AI-Enhanced SOC, including the time of the first detection and the ultimate containment. The greater success of the AI-Enhanced SOC was especially observed in its management of unknown threats. According to the Zero-Day Event Response Timeline (Table 10), the Traditional SOC was able to contain the anomaly in 42 min with the 38-min window on the manual investigation focussed on hypothesis development. At this point, however, the AI-Enhanced SOC used agentic clustering and pattern-similarity analysis to detect the threat and produce a confidence-scored remediation recommendation in minutes. The 9-min mark was a successful containment time, which was a 76% reduction in time to containment. This finding underlines the pattern-adaptive reasoning ability of the agentic AI in new threatening situations.

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Table 10. Zero-day event response comparison.

Response stage	Traditional SOC	AI-Enhanced SOC	Time reduction
Alert Detection	0 min	0 min	—
Initial Analysis	0–38 min	0–4 min	89%
Remediation Planning	38–40 min	4–6 min	—
Containment Execution	42 min	9 min	79%
Total Response Time	42 min	9 min	76%

Table 11. Escalation rate by severity level.

Severity	Escalation rate (%)
Low	~13.6
Medium	~14.1
High	~14.9
Critical	~13.2

Table 10's results prove the high adaptive abilities of the agentic system. Although the Traditional SOC took 38 min in the preliminary analysis and hypothesis construction, the AI-Enhanced SOC took only 4 min and 89% of the analysis time half. The manual workflow took 42 min to have an AI system fully contained, unlike 9 min. This 76% total time savings demonstrates that agentic AI is capable of operating successfully during high-uncertainty incidents that naturally overwhelm human-centric operations.

A case of escalation behaviour based on the severity of the incidents was analysed to know the patterns of decision making. Table 11 shows that the escalation rates were rather similar (ranging between 13.2 and 14.9) whether an alert was considered as Low or Critical. This non-correlation implies that in conventional workflows, the escalation decisions are frequently influenced by the subjective analysis of analysts instead of a prioritisation system, which again indicates the necessity of the objective and data-driven approach that agentic AI can offer.

Further investigation of the detection effectiveness was done by looking at the ratio of false positives in the various alert categories. False-positive behaviours are quite different and as it is summarised in Table 12, the highest ratio is the highest (approximately 0.35) in Policy Violations and the lowest (approximately 0.08) in

Table 12. False-positive ratio by alert category.

Alert category	False-positive ratio
Policy Violation	~0.35
Phishing	~0.23
Malware	~0.17
Beaconing	~0.12
Lateral Movement	~0.08

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Lateral Movement. The outcome logs of the remediation show that the agentic SOC was more accurate in closing these categories, and less reverse occurrence after the remediation. This implies that agentic decision logic is not only more effective in terms of accurately detecting the attack but is also better in selecting the most fitting response.

Remediation outcome logs further indicate higher closure accuracy in the agentic SOC, with fewer post-remediation reversals and analyst overrides. This suggests that agentic decision logic improves not only detection precision but also response appropriateness.

To provide a rigorous technical substantiation of the agentic AI's performance, a confusion matrix was generated based on the 90-day telemetry data $N = 5{,}571$ events. The matrix evaluates the model's ability to distinguish between actual security threats and benign noise.

The classification performance is mathematically defined by the following metrics:

- Precision (Positive Predictive Value): Reflects the probability that an alert flagged by the AI is an actual incident, minimising false alarms.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{1240}{1240 + 180} \approx 0.873.$$

- Recall (Sensitivity): Measures the system's ability to capture all true threats within the environment.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{1240}{1240 + 95} \approx 0.929.$$

- F1-Score: The harmonic mean of Precision and Recall, used to ensure the model maintains a balance between sensitivity and accuracy.

$$F1 - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{1240}{1240 + 95} \approx 0.900.$$

- Accuracy: The overall correct classification rate across all telemetry.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{1240 + 4056}{5571} \approx 0.951.$$

As illustrated in Fig. 10, the system identifies 4,056 True Negatives (98.8% of benign noise). This high TN rate is the technical driver behind the "Experience" dimension of the 3E framework, as it represents the volume of alerts successfully suppressed to prevent analyst fatigue. The True Positive (TP) count of 1,240 validates the "Effectiveness" dimension, demonstrating a high rate of successful autonomous remediations.

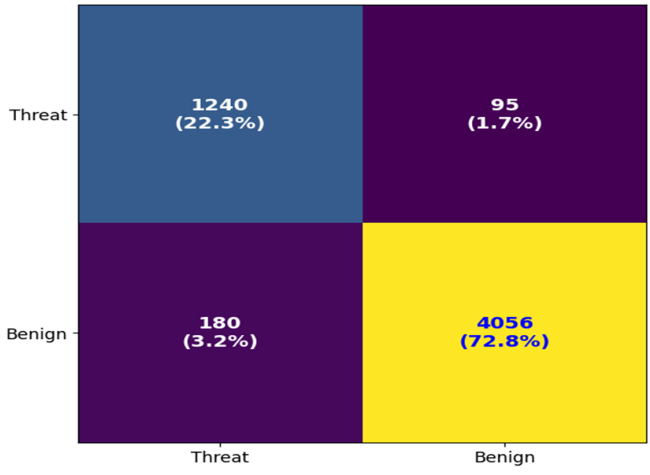


Fig. 10. Confusion matrix: Agentic SOC performance (N=5,571).

5. Discussion

This research paper will attempt to assess operational and organisational effects of agentic AI in SOC with a multi-dimensional assessment framework. Using the findings as a mapping to the proposed 3E model and the stated research questions, the discussion can explain the results as a sign of a wider socio-technical change instead of a simple performance optimisation. The findings that answer RQ1 show that agentic AI is significant in terms of operational efficiency, which is evidenced by decreases in MTTD and Mean Time to Respond. The importance of these improvements is not only about increased speed of performance, though. The fact that the response-time variance has been reduced is an indication that agentic decision-making corrects the operations of SOC when there is a change in the load of alerts. Theoretically, this observation is consistent with the socio-technical systems theory that focusses on the resilience and predictability of the system as a measure of operational maturity. Conventional SOCs are volatile in performance due to the availability of analysts, cognitive overload and workload saturation. Conversely, agentic SOC architecture separates the continuity of operation and the capacity of the individual analysts to scale performance to a consistent level. These results indicate that the efficiency of AI-enabled SOCs cannot be viewed only as speed, but as reliability in operations on a scale. These reframing issues are present with SOC benchmarking practices where the focus is on average response times and variance, and tail latency is not considered. To answer RQ2, the fact that the false-positive rates shifted significantly in the agentic SOC is important because it indicates a change in the effectiveness of detection due to increased alerting of volumes provided by 3D volume and reduced alerting through context-sensitive signal generation. This is per the previous criticism of SIEM-based detection models, which focus on sensitivity at the cost of accuracy. AI systems that are agentic use

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contextual correlation and historical learning to filter out non-actionable alerts to achieve a better signal-to-noise ratio. Notably, such reduction is not observed to undermine remediation results, as reflected in reduced post-remediation reversals and analyst overrides. Evaluation-wise, the results indicate that detection measures need to be assessed based on the quality of decisions made other than the number of alerts. To a large degree, this difference is not reflected in any of the current SOC maturity models where detection accuracy is considered constant technical properties and not an adaptive operational consequence. The answers to RQ3 show that there is a reorganisation of analyst functions in agentic SOC environments. Instead of removing human intervention, agentic AI repatriates cognitive work in place of repetitive triage processes to supervisory, interpretative, and governance work. This role change is in line with the human–AI collaboration theory that advocates complements but not substitutes. SOCs can reduce alert overload and frequency of escalation, thereby alleviating cognitive fatigue and allowing analysts to perform activities that are more valuable and need contextual judgment and strategic reasoning. Nevertheless, new skills are also brought about by this change. The analysts need to acquire AI oversight, decision auditing, and exception handling competencies. Analyst experience, therefore, should not be considered solely through the reduction of workload, but it must also be looked at considering the emerging skills requirements and professional identity as far as SOC operations are concerned. The findings on RQ4 also indicate that the advantages of autonomy do not cut across all SOC functions in the same manner. The most significant effect of agentic AI can be seen in the situation of high-volume and low-context, routine policy violations, and similar repetitive types of alerts. On the other hand, high-severity and ambiguous-context incidents that may affect the business still need human judgment. This observation supports the idea of limited autonomy, where the power of decisions is assigned to AI systems under regulated conditions of operational and governance. Instead of pursuing complete automation, successful SOC autonomy can be a layered model where AI is applied in making more mundane decisions, but humans maintain strategic control. The continuity of human governance also covers the issues of ethicality and accountability relating to autonomous security operations. The agentic SOC architecture allows alignment between efficiency and organisational accountability by ensuring that critical decisions are made by humans in the loop. The findings indicate that agentic AI is transformative and is poorly represented in existing SOC maturity models. The classic models of maturity consider it as a transition between manual operations and automated procedures, and they do not take into consideration the autonomy of decisions, the ability to learn, and the reconfigurations of human–AI roles. The 3E framework presented provides a different prism in which the accomplishment of SOC may be assessed. The framework embodies both organisational adaptation and technical performance since it incorporates efficiency, effectiveness, and experience. This combined vision is specifically applicable to the SOCs changing their tool-focussed optimisation to intelligence-directed operations. Although the

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advantages have been shown, the results also shed light on the reasons why most organisations have a hard time going beyond partial automation. The obstacles are divided into disjointed data telemetry, outdated data format, lack of confidence in AI decision-making, and organisational avoidance of giving authority. These obstacles imply that the effective implementation of agentic SOC-based systems needs to be accompanied by technical maturity as well as organisational balance, governance systems and cultural transformation. It is important to consider agentic AI as a socio-technical system, and not as an independent tool, when it comes to sustainable transformation. In general, the discussion shows that agentic AI presents a qualitative change in the operations of SOC. Manifold business advantages in the form of stability in operations, an improvement in effectiveness in the form of improved signal quality, and a redefinition of the roles of analysts. These transformations back the claim that autonomy and not automation is the next phase of evolution of SOC.

6. Conclusion

The development of SOCs is being more determined not by the quantity of tools placed into it, but by how much decision authority organisations will grant intelligent systems. The paper will contend that the shift in the mode of the operation of the SOCs, moving from manual and script-based operation toward the agentic operation of the AI-enabled architecture, is a structural change, not merely an efficiency improvement. This work offers a comprehensive perspective for evaluating SOC performance in a setting where autonomy, learning, and collaboration between humans and AI are co-presented. The results show that agentic AI transforms the logic of operation of SOCs by scaling up performance, enhancing signal quality, and altering the role of the analysts, who are transformed to perform supervisory governance instead of execution-based work. The changes uniquely indicate that autonomy serves as a trigger in SOC maturity, which helps organisations to handle complexity that is in the way of human-only operation. Besides the operational advantages that are short-term in nature, the study is able to define the strategic implications of agentic SOC adoption. Along with the progress in the decision-making process towards intelligent systems, governance, calibration of trust and accountability become paramount towards sustainable deployment. Organisations that consider agentic AI as a socio-technical complex, i.e. a combination of technical functionality and human governance, organisational preparedness, are more likely to achieve long-term value. In the future, an upgrade of human analysts into judgment, validation, and strategic risk management will not be used to describe the future SOC, but rather the absence of human analysts. AI systems will progressively become agentic to take on decision cycles occurring over routine and large volumes, leaving human expertise to focus on situations that require contextual thinking and ethical decisions. The given research provided the basis on which future research on autonomous security operations can be

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conducted, as it offers a systematic review framework and empirical-based knowledge. Mechanisms of explainable AI should be integrated, longitudinal deployments should be investigated, and human control should be refined further in future work. With cyber threats ever growing in size and complexity, the capacity to produce, assess and manage intelligent SOC models will become an intrinsic aptitude of robust organisations.

According to an IKM approach, agentic SOCs reorganise organisational bodies of knowledge. Creation of knowledge is active and pattern-directed; classification is systematic, experience of past undocumented knowledge is captured by classifying it, and learning is ongoing, with every event making improvements to the collective intelligence. These changes provide instructions to any field of knowledge-intensive work where AI takes the decision-making role and humans control regulation.

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